Sen Fang

(+1) 984-379-1827
sfang9@ncsu.edu
Ph.D. Candidate in Computer Science — NC State University tomasandersonfang.github.io · Google Scholar

RESEARCH FOCUS

I build **intelligent systems that automatically detect software vulnerabilities** by combining large language models with program analysis. My work bridges deep learning and static analysis through **LLM-based agents** that reason over code structure (Code Property Graphs, ASTs) and autonomously wield developer tools (CodeQL, Joern, grep, bash) to find, verify, and explain real-world security flaws. I also investigate **LLM security and robustness** in code-related tasks, ensuring the models we deploy are themselves trustworthy.

Education

NC State University — Aug 2024 – Present
Ph.D. in Computer Science · Advisor: Prof. Bowen Xu (SoftMax Lab)
Focus: LLM-driven vulnerability detection, AI agents for software security, LLM robustness

Central China Normal University — Sep 2018 – Jun 2020
M.Sc. in Electronics & Communication Engineering · Advisor: Prof. Shaocheng Qu

Wuhan Polytechnic University — Sep 2014 – Jun 2018
B.Sc. in Electronic Information Engineering Outstanding Student 2018

Research & Industry Experience

- **Hedra** — May 2025 – Aug 2025
  Machine Learning Engineer Intern · Supervised by Hongwei Yi
  Led development of a petabyte-scale video data processing pipeline, curating millions of high-quality video samples for upstream post-training of generative models (text-to-video, image-to-video).

- **KTH Royal Institute of Technology** — Mar 2023 – Mar 2024
  Research Engineer · Advisor: Prof. Martin Monperrus
  Developed RepairLLaMA (LoRA-based program repair) and Supersonic (LLM-driven C/C++ optimization), both published in IEEE TSE. Contributed to AI-driven test data generation research.

- **Macau University of Science and Technology** — Sep 2020 – Nov 2022
  Research Assistant · Advisor: Prof. Tao Zhang
  Led research on learning-based bug report understanding, code clone detection, and pull request automation. Published 5 papers across ICSE, IEEE TSE, IEEE TR, JSS, and IST.

Selected Publications

15 PAPERS **400+** CITATIONS **8** FIRST / EQUAL-FIRST AUTHOR

VULNERABILITY DETECTION & SOFTWARE SECURITY

[1] **S. Fang**, W. Ding, B. Xu. "AEGIS: Multi-Agent CPG-Augmented Framework for Automated Vulnerability Detection." *Submitted to ACM SIGPLAN OOPSLA, 2026.* Under Review

[2] Y. Huang, **S. Fang**, J. Li, J. Tao, B. Hu, T. Zhang. "Deep Smart Contract Intent Detection." *IEEE SANER 2025.* CCF-A

[3] Y. Li, **S. Fang**, et al. "Enhancing Android Malware Detection: The Influence of ChatGPT on Decision-centric Tasks." *ACM TOSEM, 2025.* CCF-A*

LLM ROBUSTNESS

[4] **S. Fang**, W. Ding, A. Mastropaolo, B. Xu. "Smaller = Weaker? Benchmarking Robustness of Quantized LLMs in Code Generation." *IEEE TSE.* Major Revision CCF-A*

[5] **S. Fang**, W. Ding, B. Xu. "EVALOOP: Assessing LLM Robustness in Programming from a Self-consistency Perspective." *ACM TOSEM.* Major Revision CCF-A*

[6] Y. Li, T. Zhang, X. Luo, H. Cai, **S. Fang**, D. Yuan. "Do Pre-trained Language Models Indeed Understand Software Engineering Tasks?" *IEEE TSE.* CCF-A*

AUTOMATED PROGRAM REPAIR & CODE OPTIMIZATION

[7] A. Silva*, **S. Fang***, M. Monperrus. "RepairLLaMA: Efficient Representations and Fine-Tuned Adapters for Program Repair." *IEEE TSE, 2025.* CCF-A* (*equal contribution)

[8] Z. Chen, **S. Fang**, M. Monperrus. "Supersonic: Learning to Generate Source Code Optimizations in C/C++." *IEEE TSE, 2024.* CCF-A*

[9] B. Baudry, K. Etemadi, **S. Fang**, et al. "Generative AI to Generate Test Data Generators." *IEEE Software.*

MINING SOFTWARE ENGINEERING & CODE INTELLIGENCE

[10] **S. Fang**, T. Zhang, Y. Tan, H. Jiang, X. Xia, X. Sun. "RepresentThemAll: A Universal Learning Representation of Bug Reports." *ICSE 2023.* CCF-A*

[11] D. Yuan*, **S. Fang***, T. Zhang, Z. Xu, X. Luo. "Java Code Clone Detection by Exploiting Semantic and Syntax Information from Intermediate Code-Based Graph." *IEEE TR, 2022.* CCF-A (*equal contribution)

[12] **S. Fang**, T. Zhang, Y. Tan, Z. Xu, Z. Yuan, L. Meng. "PRHAN: Automated Pull Request Description Generation Based on Hybrid Attention Network." *JSS, 2022.* CCF-A

[13] **S. Fang***, Y. Tan*, T. Zhang, Z. Xu, H. Liu. "Effective Prediction of Bug-Fixing Priority via Weighted Graph Convolutional Networks." *IEEE TR, 2021.* CCF-A (*equal contribution)

[14] **S. Fang**, Y. Tan, T. Zhang, Y. Liu. "Self-Attention Networks for Code Search." *IST, 2021.* CCF-A

[15] Y. Tan, J. Chen, W. Shang, T. Zhang, **S. Fang**, X. Luo, Z. Chen, S. Qi. "STRE: An Automated Approach to Suggesting App Developers When to Stop Reading Reviews." *IEEE TSE.* CCF-A*

Teaching

Teaching Assistant — CSC 326: Software Engineering
NC State University, Fall 2024 & Spring 2025

Teaching Assistant — CSC 484/584: Building Gaming AI
NC State University, Spring 2025

Professional Service

Peer Reviewer: IEEE TSC, ACM TOSEM, IST, EAAI, JSS, ASE, AIR, JCC, and others

Honors & Awards

Outstanding Student Award
Wuhan Polytechnic University, 2018

Technical Expertise

VULNERABILITY & PROGRAM ANALYSIS
Joern, CodeQL, SpotBugs, FindSecBugs, Code Property Graphs, AST / CFG / DFG analysis

LLM TRAINING & INFERENCE
PyTorch, Transformers, LoRA / QLoRA / PEFT, vLLM, DeepSpeed, SLURM, JAX

AGENT & RETRIEVAL SYSTEMS
Multi-agent orchestration, RAG pipelines, tool-augmented LLM agents

LANGUAGES & PLATFORMS
Python, C/C++, Java, LaTeX · Linux, Jupyter, Cursor, Claude Code